

# Raj Kumar Nelluri

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## PROFESSIONAL SUMMARY

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AI / ML Engineer building production-grade intelligent systems combining machine learning, LLMs, and deterministic decision engines. Experienced in designing end-to-end pipelines from data ingestion to deployed inference, with a focus on explainable and auditable AI systems over black-box models. Built and deployed scalable applications on AWS including financial decision engines, enterprise RAG systems, and predictive ML pipelines.

## TECHNICAL SKILLS

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**Programming:** Python, SQL

**AI / LLM Systems:** Retrieval-Augmented Generation (RAG), LLM Applications, Deterministic Decision Systems, AI System Design

**Machine Learning:** Scikit-learn, XGBoost, TensorFlow

**LLM & AI Tools:** OpenAI API, LangChain, Prompt Engineering, Embeddings, Vector Databases

**Backend & APIs:** FastAPI, REST APIs

**Cloud & Deployment:** AWS (EC2, SageMaker, Lambda, S3), Docker, Redis

**Data & Analytics:** Pandas, NumPy, Matplotlib, Tableau, Power BI

## AI ENGINEERING PROJECTS

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### AI Financial Research Agent | [Live Demo](#) | [GitHub](#)

2026

*FastAPI | GPT-4o | Alpha Vantage | Redis | AWS EC2 | Docker*

- Built a production-grade financial analysis system integrating technical indicators, company fundamentals, and news sentiment into a deterministic multi-factor scoring engine
- Designed normalized scoring with dynamic max adjustment, eliminating missing data bias and reducing false SELL signals from ~100% to threshold-based outcomes
- Developed conflict-aware decision logic detecting disagreement across signals and defaulting to HOLD in ambiguous scenarios to avoid false conviction
- Implemented portfolio allocation engine ranking multiple stocks and distributing capital proportionally based on normalized scores
- Architected and deployed a 7-step async pipeline on AWS EC2 with FastAPI and Redis caching (15 min TTL), reducing repeated query latency from ~ 15s to <100ms
- Integrated GPT-4o strictly for explainability (sentiment classification and reasoning), ensuring all decisions remain deterministic and auditable
- Eliminated reliance on LLM-based decision making by constraining GPT-4o to explanation-only role, ensuring fully reproducible and auditable outputs

### Enterprise RAG Chatbot – LLM-Powered Document Intelligence | [Live Demo](#) | [GitHub](#)

2026

*Python | LangChain | OpenAI | ChromaDB | Docker | AWS EC2*

- Designed and deployed a Retrieval-Augmented Generation (RAG) system enabling users to upload and query PDF documents with citation-grounded responses
- Architecture: PDF → Chunking → Embeddings → ChromaDB → MMR Retrieval → LLM
- Built semantic retrieval pipeline using ChromaDB vector database with Maximum Marginal Relevance (MMR) search and reranking to improve answer relevance
- Implemented hallucination mitigation using similarity-threshold filtering and context-grounded prompts
- Containerized application using Docker and deployed on AWS EC2 with persistent vector storage
- Achieved ~ 2 second end-to-end response latency including retrieval and LLM generation

### Customer Churn Prediction Pipeline | [GitHub](#)

2025

*Python | XGBoost | AWS SageMaker | Lambda | CloudWatch*

- Built end-to-end ML pipeline on AWS for customer churn prediction including S3 ingestion, Lambda ETL orchestration, and SageMaker model training
- Deployed XGBoost model as SageMaker real-time inference endpoint and implemented automated monitoring with CloudWatch and EventBridge

## ADDITIONAL PROJECTS

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| <b>Retail Sales Forecasting System</b>   <a href="#">GitHub</a>   <i>Python, XGBoost, Flask</i>                                                                                                                                                                                                                                                                                                            | 2025 |
| <ul style="list-style-type: none"><li>• Built time-series forecasting model using XGBoost to predict retail demand from 500,000+ transactions</li><li>• Engineered lag features, rolling averages, and seasonal indicators improving demand prediction accuracy</li></ul>                                                                                                                                  |      |
| <b>Insurance Fraud Detection System</b>   Python, Scikit-learn, AWS Lambda, Kinesis, RDS                                                                                                                                                                                                                                                                                                                   | 2024 |
| <ul style="list-style-type: none"><li>• Developed fraud detection pipeline processing 15,000+ insurance claims using Random Forest and Gradient Boosting models</li><li>• Implemented real-time data ingestion using AWS Kinesis and serverless processing with Lambda</li><li>• Built ETL validation pipeline and stored fraud scores in AWS RDS enabling analysts to query predictions via SQL</li></ul> |      |

## EDUCATION

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|----------------------------------------------------------|----------------------------|
| <b>Pace University</b>                                   | New York City, NY          |
| <i>Master of Science in Computer Science</i>             | <i>Sep 2023 – Apr 2025</i> |
| <b>Amrita Vishwa Vidyapeetham</b>                        | Bengaluru, India           |
| <i>Bachelor of Technology in Artificial Intelligence</i> | <i>Jul 2019 – Apr 2023</i> |

## CERTIFICATIONS

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|---------------------------------------------------------------|------|
| <b>AWS Certified Cloud Practitioner</b> – Amazon Web Services | 2025 |
| <b>Adobe Journey Optimizer Foundation</b> – Adobe             | 2025 |
| <b>Complete Python Developer</b> – Udemy                      | 2023 |